

BHADRINATHAN A

Software Engineer | Technical Lead

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SUMMARY

Software engineer with 4 years of experience shipping full-stack products end to end — from a police records CRM to cloud lab infrastructure and an edtech marketplace, and most recently production Retrieval-Augmented Generation (RAG) and agentic AI systems. Author of fastapi-querybuilder, an open-source Python package with 16,000+ PyPI downloads. Progressed from Software Developer to Technical Lead at BloomSkillTech, owning architecture, engineering processes, and delivery.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Go, SQL, Rust (basic)
- **Backend:** FastAPI, REST APIs, SQLAlchemy, Server-Sent Events, PostgreSQL, Redis
- **AI / LLM:** Retrieval-Augmented Generation (RAG), LangChain, LangGraph, pgvector, LLM evaluation, AI agents
- **Frontend:** React, Next.js, Tailwind CSS
- **Infrastructure & DevOps:** Docker, Linux, GitHub Actions (CI/CD), Nginx, WireGuard VPN, container orchestration
- **Practices:** System design (HLD/LLD), code review, automated testing, phased delivery, agentic AI coding tools

EXPERIENCE

BloomSkillTech — Salem, Tamil Nadu (Hybrid)

Technical Lead - EdTech Platform

Jan 2025 – Present

- Led the end-to-end build of a two-sided edtech marketplace connecting trainers with institutions — from system architecture and data modelling through production launch.
- Directed engineering across the React/Next.js frontend, FastAPI/PostgreSQL backend, and cloud infrastructure.
- Established team engineering processes: CI/CD pipelines, automated testing, code reviews, and phased delivery.
- Owned the technical roadmap, balancing delivery speed, infrastructure cost, and long-term scalability.

Software Developer

May 2023 – Dec 2024

- Built a cloud labs platform that provisions isolated, browser-accessible development environments on demand, removing all local setup for users.
- Developed the core full-stack product: user-facing frontend, backend APIs, and environment provisioning services, with containerized labs for consistent, reproducible sessions.
- Designed and implemented a secure WireGuard VPN layer for private, per-session network access to lab environments, with controlled resource access and isolation.
- Built the orchestration layer to automatically provision, scale, and tear down containerized environments, optimizing cloud resource usage and cost.
- Drove code quality and architectural direction through reviews and guidance across the platform.

Cyber Crime Police Station — Salem, Tamil Nadu

Software Engineer Intern

Mar 2022 – Mar 2023

- Sole developer of a complete CRM that digitized crime records with end-to-end case timeline tracking, enabling officers to log, search, and monitor investigations.
- Built with a React frontend, Go backend APIs, and PostgreSQL; containerized and deployed with Docker.

OPEN SOURCE & PROJECTS

- **fastapi-querybuilder** — authored Python package adding filtering, sorting, and search to FastAPI + SQLAlchemy (14 operators, nested relationship joins, pagination, soft-delete); 16,000+ PyPI downloads, MIT.
- **fastapi_sse_events** — published PyPI library that simplifies Server-Sent Events for FastAPI, backed by Redis pub/sub for horizontal scale.
- **ZeroVPN** — self-hosted WireGuard VPN manager: a multi-crate Rust (Axum) backend and React frontend sharing one schema over WASM + MessagePack.
- **RAG Knowledge System** — Retrieval-Augmented Generation with pgvector semantic search, LangGraph agent orchestration, and an LLM evaluation harness.
- **Docker Stats Monitor** — real-time web dashboard streaming container status, CPU/memory usage, and network/block I/O over WebSockets.

EDUCATION

B.E. Electronics and Communication Engineering, Muthayammal Engineering College

2019 – 2023